Marien Chenaud

PhD Candidate & Research Scientist | Scientific Machine Learning | Digital Twins

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Selected Research Contributions

- Developed physics-informed mesh-based graph methods that combine graph neural networks with finite-element numerical kernels to solve partial differential equations on complex 2D/3D geometries, leading to a peer-reviewed journal publication in *Advances in Engineering Software*.
- Identified limitations of automatic-differentiation-based PDE residuals on complex geometries and proposed solver-compatible residual constructions for physics-informed learning.
- Designed geometry-preserving hierarchical connectivity mechanisms to improve information flow in graph neural networks for scientific machine learning; presented the work at an international conference and extended it into a journal manuscript.
- Investigating numerical and pointwise error estimation for physics-informed neural networks, with manuscripts under review/in preparation on certification and reliability of learned partial differential equation solvers.
- Contributed to the development of a Python library interfacing Fortran-based finite element numerical operators, enabling their integration into machine learning workflows.

Technical Skills

Research areas: Scientific machine learning, deep learning-based surrogate models, graph neural networks, physics-informed learning, neural operators, numerical analysis, finite-element methods, optimisation, statistical learning, computational fluid dynamics, partial differential equations, digital twins, computer vision.
Methods: Mesh-based graph models, geometry-preserving hierarchical connectivity, physics-informed objectives, hybrid numerical/ML solvers, inverse problems, RANS simulation, U-Nets, convolutional neural networks.
Programming: Python (PyTorch, NumPy, TensorFlow, JAX), Fortran, R, FreeFem++.
Software and tools: Git, LaTeX, Inkscape, Linux.

Research and Professional Experience

Research Scientist — *November 2022 – Present*
Transvalor SA — Paris, France

- Develop end-to-end research projects on hybrid numerical/deep learning methods for multiphysics, spatiotemporal industrial applications, from mathematical modeling and literature review to implementation, validation, and publication.
- Build models for physical systems surrogate modeling using graph neural networks, neural operators, and physics-informed objectives, with emphasis on suitable inductive bias for generalisation to complex geometries and physically consistent behaviour.
- Develop Python/PyTorch and Fortran code connecting learned models with existing numerical simulation workflows for mechanical processes and digital-twin prototypes.
- Evaluate learned solvers across accuracy, stability, physical consistency, robustness to geometry/mesh changes, and compatibility with finite-element-based numerical kernels.
- Collaborate with computational physicists, machine-learning researchers, and engineering-software teams to translate industrial simulation requirements into research questions and model development.

Research Intern, Machine Learning for Fluid Mechanics — *April 2021 – August 2021*
CEA Saclay — Saclay, France

- Studied coupling strategies between machine learning models and numerical simulations for computational fluid dynamics, with emphasis on Reynolds-averaged Navier–Stokes simulations.
- Built and assessed proof-of-concept ML/simulation workflows for fluid-mechanics use cases; presented the work as a poster at the TrioCFD Seminar.

Research Intern, Numerical Analysis — *September 2020 – January 2021*
CentraleSupélec Laboratory — Gif-sur-Yvette, France

- Studied partial differential equation-based methods for cancer tumour detection, combining finite-element numerical simulation with inverse-problem formulations.
- Implemented numerical experiments for forward resolution and inverse parameter estimation in applied biomedical modelling.

Data Analysis Consultant *2020*
Paris Public Transportation Company / CentraleSupélec Student Consulting Contract Paris, France

- Conducted a statistical study of a metro wear component for an industrial maintenance use case.
- Delivered data-analysis findings and recommendations under a student consulting contract with the RATP company.

Publications and Manuscripts

- Chenaud, M., Anwer, N., Alves, J., & Magoulès, F. (2026). *Pointwise error estimates for numerical physics-informed neural networks*. In preparation.
- Chenaud, M., Alves, J., & Magoulès, F. (2026). *A numerical approach to physics-informed learning for non-analytic problems*. Submitted to *Journal of Computational Physics*. Under review.
- Chenaud, M., Alves, J., & Magoulès, F. (2026). *Enhancing information flow in graph neural networks through geometry-preserving hierarchical connectivity*. Submitted to *Computers & Structures*. Under review.
- Chenaud, M., Alves, J., & Magoulès, F. (2026). *How to learn Physics: A practical review of scientific machine learning*. *Archives of Computational Methods in Engineering*. DOI / pdf
- Anwer, N., Chenaud, M., & Elizondo, D. (2026). *Certificates for complex-compatible learned cochain Laplacians*. *International Conference on Machine Learning (ICML)*, main track. Accepted.
- Chenaud, M., Alves, J., & Magoulès, F. (2025). *Enhancing information flow in graph neural networks for scientific machine learning*. *The Seventh International Conference on Artificial Intelligence, Soft Computing, Machine Learning and Optimization in Engineering*. DOI / HAL.
- Chenaud, M., Magoulès, F., & Alves, J. (2024). *Physics-Informed Graph-Mesh Networks for PDEs: A hybrid approach for complex problems*. *Advances in Engineering Software*, 197, 103758. DOI / arXiv.
- Chenaud, M., Alves, J., & Magoulès, F. (2023). *Physics-Informed Graph Convolutional Networks: Towards a generalized framework for complex geometries*. *The Sixth International Conference on Soft Computing, Machine Learning and Optimization in Civil, Structural and Environmental Engineering*. DOI / arXiv.

Education

PhD Candidate, Applied Mathematics / Scientific Machine Learning *2022 – Expected 2026*
CentraleSupélec, Université Paris-Saclay
Thesis: Hybrid Scientific Machine Learning Methods for Surrogate Modeling of Physical Systems.

- Research focus: scientific machine learning for PDEs, graph neural networks on meshes, physics-informed learning, neural operators, finite-element/numerical-solver integration, and digital twins for multiphysics, spatiotemporal processes.
- Teaching: Scientific Computing, Machine Learning, and Linear Algebra; teaching assistant for Bachelor's and Engineering degree courses at CentraleSupélec.
- Research communication: Transvalor Tech Days 2025, plenary talk, Bergamo, Italy; Transvalor International Simulation Days 2024, oral presentation, Cannes, France; TrioCFD Seminar 2021, poster presentation, CEA Saclay, France.

Engineering Degree, Mathematics and Data Science Track *2018 – 2022*
CentraleSupélec
Relevant coursework: Statistics, Machine Learning and Deep Learning, Functional Analysis, Physics, Computer Science.

Master's Degree in Mathematics of Randomness, Statistics and Machine Learning Track *2020 – 2021*
Université Paris-Saclay
Relevant coursework: Stochastic Calculus, High-Dimensional Statistical Learning, Machine Learning Project.

Semester Abroad, Erasmus Programme *2020*
NTNU University, Trondheim, Norway
Relevant coursework: Optimisation, probabilistic and statistical modelisation, climate modelisation, philosophy.

Bachelor's Degree in Fundamental and Applied Mathematics *2018 – 2019*
Université Paris-Saclay

Preparatory Classes for the Grandes Écoles, Mathematics and Physics *2016 – 2018*
Lycée Thiers, Marseille, France

Languages

French (Native) | English (Fluent, C1+) | Spanish (Intermediate)